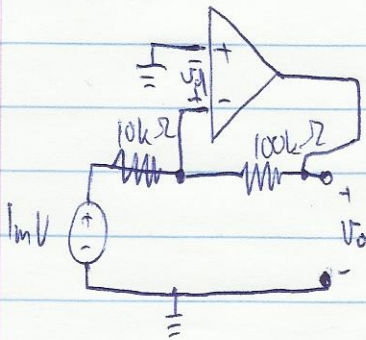


Problem 5.7

similar to example 5.1 pg 178



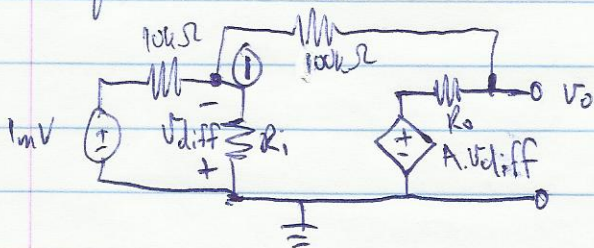
$$R_i = 100 \text{ k}\Omega$$

$$R_o = 100 \Omega$$

$$A = 100,000$$

determine v_d and v_o

Equivalent circuit:



Note from first diagram:

$$v_{diff} = -v_d$$

KCL @ node 1:

$$\frac{v_1 - 1\text{m}}{10\text{k}} + \frac{v_1 - 0}{100\text{k}} + \frac{v_1 - v_o}{100\text{k}} = 0$$

$$\Rightarrow v_1 \left(\frac{1}{10\text{k}} + \frac{1}{100\text{k}} + \frac{1}{100\text{k}} \right) + v_o \left(-\frac{1}{100\text{k}} \right) = \frac{1\text{m}}{10\text{k}} \dots (1)$$

KCL @ node 0:

$$\frac{v_o - v_1}{100\text{k}} + \frac{v_o - 100,000 v_{diff}}{100} = 0$$

$$\text{but } v_{diff} = -v_1$$

$$\Rightarrow v_1 \left(-\frac{1}{100\text{k}} + \frac{100,000}{100} \right) + v_o \left(\frac{1}{100\text{k}} + \frac{1}{100} \right) = 0 \dots (2)$$

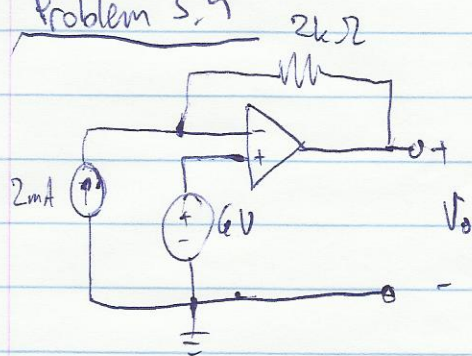
solving simultaneously:

$$v_1 = 100 \text{ nV}$$

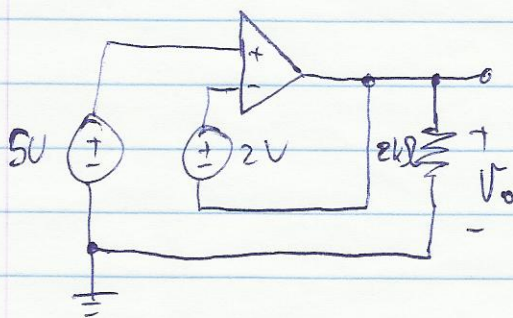
$$\therefore \underline{v_d = 100 \text{ nV}}$$

$$\underline{v_o = -9.999 \text{ mV}}$$

Problem 5.9

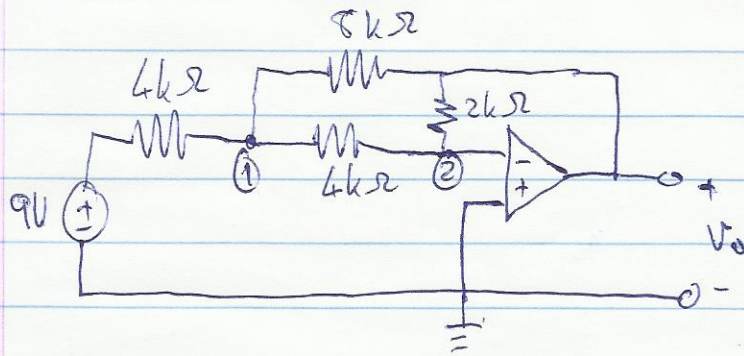


$$\begin{aligned} V_o &= 6V - (2k\Omega)(2mA) \\ &= 6 - 4 \\ &= 2V \end{aligned}$$



$$\begin{aligned} V_o &= 5V - 2V \\ &= 3V \end{aligned}$$

Problem 5.20



KCL @ node 1:

$$\frac{V_1 - 9}{4k} + \frac{V_1 - 0}{4k} + \frac{V_1 - V_o}{8k} = 0 \quad \dots (1)$$

KCL @ node 2:

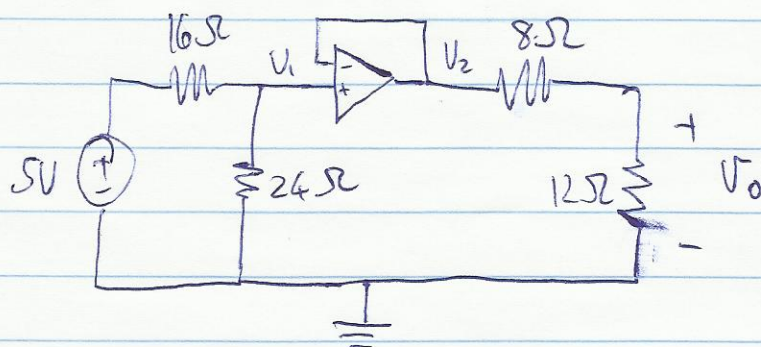
no current flows into op-amp,
node 2 is a "virtual ground."

$$\frac{V_1}{4k} = \frac{-V_o}{2k} \Rightarrow V_o = -\frac{1}{2}V_1 \quad \dots (2)$$

Solve (1) and (2) simultaneously:

$$\underline{V_o = -1.64 V}$$

Problem 5.27



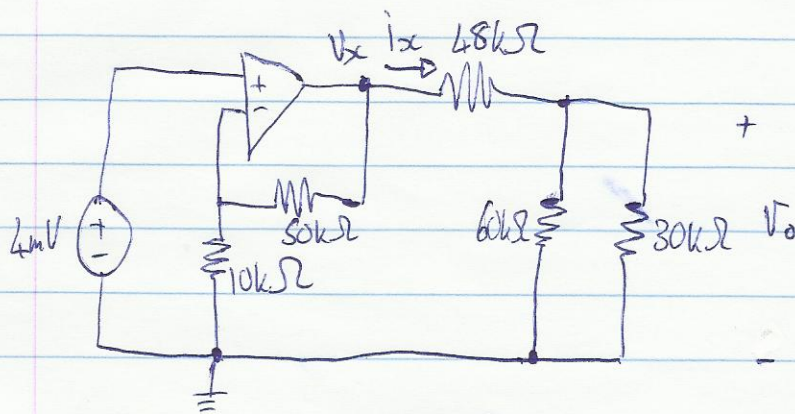
Voltage divider:
$$V_1 = \frac{24}{16+24} (5V)$$
$$= 3V$$

$$V_1 = V_2$$

Insight: The op-amp acts as a voltage buffer.

Voltage divider:
$$V_0 = \frac{12}{12+8} V_2$$
$$= 1.8V \rightarrow$$

Problem 5.32



Calculate i_x , V_o

determine power dissipated
by $30\text{ k}\Omega$ resistor.

KCL @ op-amp negative input:

$$\frac{4\text{mV}}{10\text{k}\Omega} = \frac{V_x - 4\text{mV}}{50\text{k}\Omega} \quad \therefore V_x = 24\text{mV}$$

$$i_x = \frac{V_x}{R_{eq}}$$

where R_{eq} is the equivalent resistance
of the rest of the circuit.

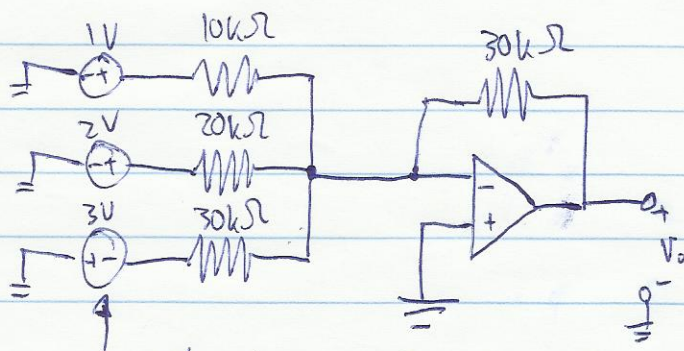
$$\begin{aligned} \therefore i_x &= \frac{24\text{mV}}{68\text{k}\Omega} \\ &= 352,9\text{ nA} \end{aligned}$$

$$\begin{aligned} R_{eq} &= 48\text{k}\Omega + (60\text{k}\Omega \parallel 30\text{k}\Omega) \\ &= 68\text{k}\Omega \end{aligned}$$

$$\begin{aligned} V_o &= \frac{20\text{k}}{68\text{k}} \cdot 24\text{mV} \\ &= 7,06\text{ mV} \end{aligned}$$

$$P = \frac{V^2}{R} = \frac{(7,06\text{ mV})^2}{30\text{k}\Omega} = 1,66\text{ nW}$$

Problem 5.37



notice the polarity of this source!

by eq. 5.15 pg 186:

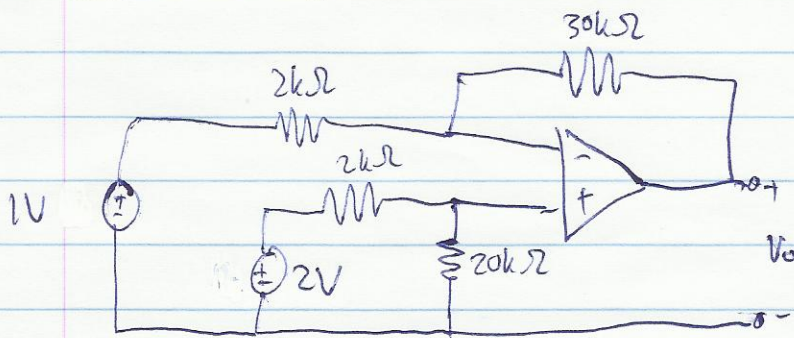
$$V_o = - \left(\frac{30k}{10k} \cdot 1 + \frac{30k}{20k} \cdot 2 - \frac{30k}{30k} \cdot 3 \right)$$

$$= -3V$$

→

determine output of summing amplifier.

Problem 3.47



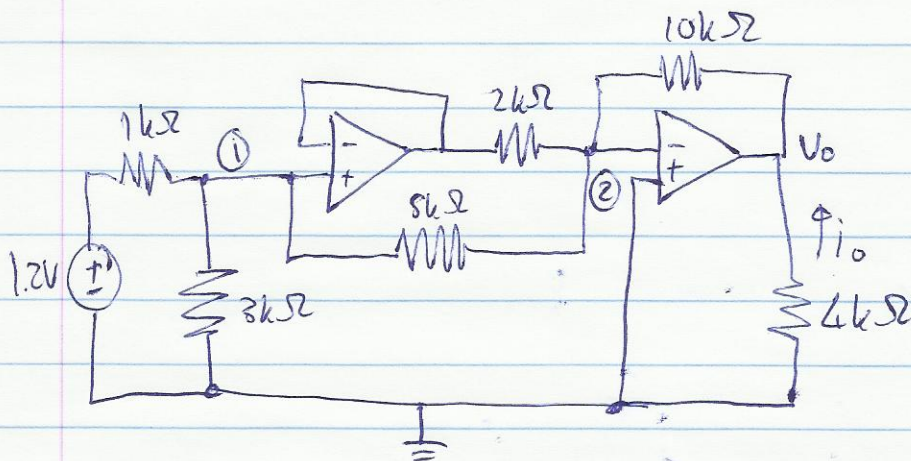
determine V_o for
the difference amplifier.

from eq. 5.18 pg 188:

$$V_o = \frac{30k \left(1 + \frac{2k}{30k} \right)}{2k \left(1 + \frac{2k}{20k} \right)} \cdot 2 - \frac{30k}{2k} \cdot 1$$

$$= \underline{14,09 \text{ V}}$$

Problem S.58



calculate i_o .

KCL @ node 1:

$$\frac{V_1 - 1,2}{1k} + \frac{V_1}{3k} + \frac{V_1 - 0}{5k} = 0$$

because V_2 is a virtual ground.

$$\Rightarrow V_1 = 782,6 \text{ mV}$$

KCL @ node 2:

$$\frac{0 - 782,6 \text{ mV}}{2k} + \frac{0 - 782,6 \text{ mV}}{5k} + \frac{0 - V_o}{10k} = 0$$

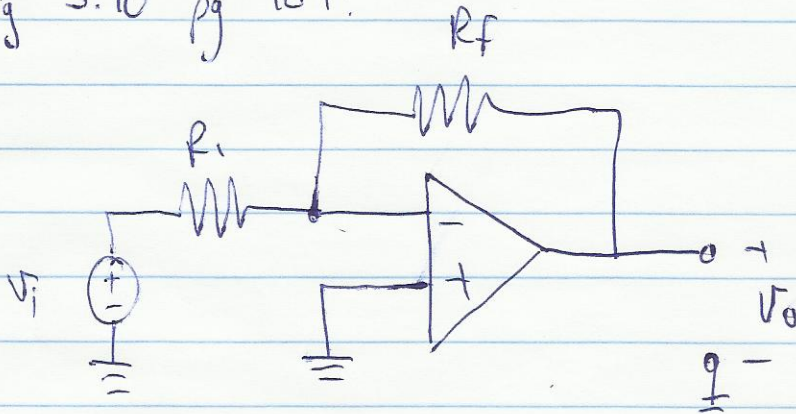
$$\Rightarrow V_o = -9,48 \text{ V}$$

$$\therefore i_o = \frac{9,48}{4k} = 1,37 \text{ mA} \rightarrow$$

Problem S.22

Design an inverting amplifier with a gain of -15 .

Fig S.10 pg 181:



eq S.9 pg 182:

$$\frac{V_o}{V_i} = - \frac{R_f}{R_i}$$

$$\therefore - \frac{R_f}{R_i} = -15$$

$$\Rightarrow R_f = 15 R_i$$

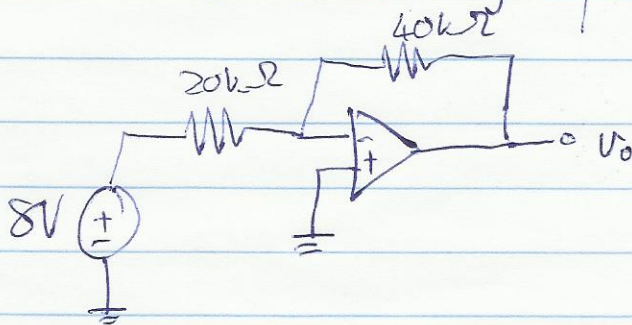
$$\text{select } R_i = 100 \text{ k}\Omega$$

$$\therefore R_f = 1,5 \text{ M}\Omega$$

} not unique. Your answer will vary based on your selection. As long as eq. S.9 holds, the answer will be correct.

Problem 5.56

Consider first the inverting amplifier:

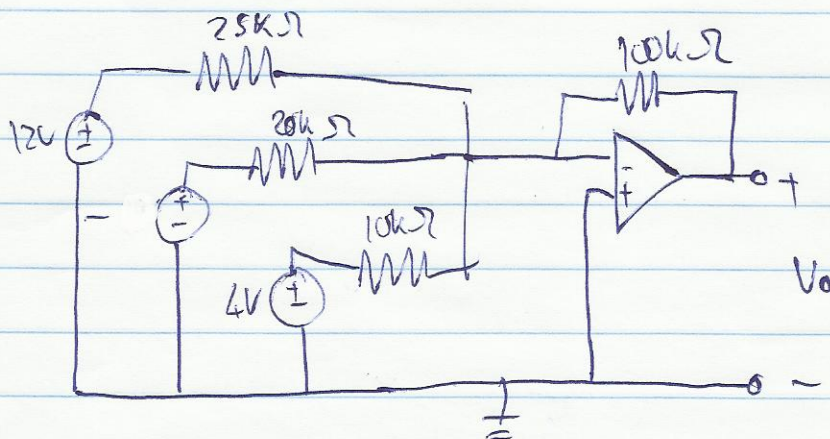


by inspection:

(eq. 5.9 otherwise)

$$V_o = -16 \text{ V}$$

Now equivalent circuit:

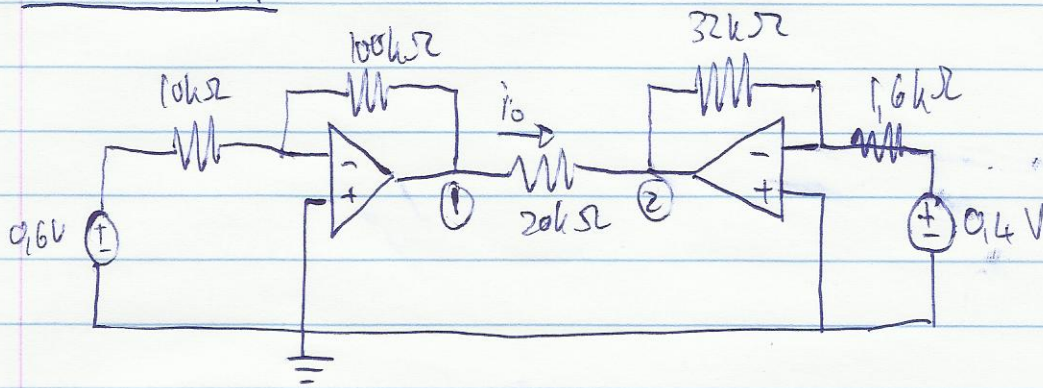


determine V_o :

eq. 5.15 pg 186:

$$V_o = - \left(\frac{100k}{25k} \cdot 12 - \frac{100k}{20k} \cdot 16 + \frac{100k}{10k} \cdot 4 \right)$$
$$= -8 \text{ V}$$

Problem 5.74



Both op-amps act as inverting amplifiers.

$$V_1 = -10(0.6) \\ = -6V$$

$$V_2 = -20(0.4) \\ = -8V$$

$$i_o = \frac{V_1 - V_2}{20k\Omega} \\ = \frac{2V}{20k\Omega} \\ = \underline{0.1mA} \rightarrow$$